

COMPUTER SCIENCE TRIPOS Part IB – 2022 – Paper 5

3 Computer Networking (awm22)

Buffers Queues
and Schedulers,
TCP, connections
and
connectionless

- (a) Using an explanation of the difference between flow-control and congestion-control, discuss the impact of a stable end-to-end latency. [5 marks]

Answer: flow control : behaviour between receiver and sender
network congestion control is a global (minimization/control) problem/optimization (1 mark each)

a stable end to end latency permits fine-granularity of flow-control; otherwise the sending system will always need some idea of the capacity of the receiver and be able to accommodate the worst case delay in flow-control stable e2e latency enables a smaller worst-case delay and thus a 'tighter/finer' control as there is a reduced need for accommodating the worst case with the transmit side accommodating the receiver side capacity to receive. (2 marks)

congestion implicit implies (blind or explicit) coordination among senders on the sharing of contested resource.

this in turn requires a congestion signal

so I'd anticipate for CompNet where classic TCP is the main discussion congestion signalling, discussion about timeouts

- (b) Mobile end-points present a challenge for any connection-based application on an IP network. Using an example, discuss how this challenge might be resolved without a complete reworking of the end-system application. [5 marks]

Answer:

Whilst other connection centricity is discussed I expect students to talk of IP and TCP over it. Clearly the underlying network being 'mobile and flexible' is one approach - essentially an entire system is used (cell phone network) to maintain the illusion the mobile device retains a static IP allocation.

I'd hope the candidate mentions time-scale, as given large movement - we group that at the "exceeds flow lifetime, so disconnect and reconnect is ok"

IP addresses mobility has been used but is complicated by the contra or independent needs for IP Addresses to identify both end-points and to be used for routing.

TCP encodes IP address data into its checksum (pseudo header) so further complicates even desirable changes to underlying IP addresses

Mobile TCP aka multipath tcp is a lovely solution - might require a little app reworking.

apps use web space addresses (and translation via CDN, anycast, URL rewriting, etc to regularly assist with movement.)

- (c) A reverse proxy such as Varnish makes static snapshots of the content of a dynamic webpage. Outline a use-case for using such a reverse proxy, paying particular attention to a case when an otherwise useful system might need disabling. [5 marks]

Answer:

There is a case to never disable Varnish, however as it implicitly provides a non-timely snapshot of dynamic content it may be only truly necessary when the target website is under heavy

workloads: one where the time taken to create dynamic content is exceeding the rate at which pages can be usefully served to clients.

- (d) In 2021 a large social network had to reboot all its systems after systemic failure. Discussing the extensive use of image caching, outline the challenges that face an image serving system in the social media space (that is, one that uses popularity, hashtags, and so-forth). Discuss in particular why, for weeks after the reboot, many users complained of poor performance, and of being served irrelevant or old images.

Hint: A distinguishing feature of social networks has been a heavy-tailed distribution of interest among images and users.

For your information: a large image sharing site has 500×10^6 active daily users, 500×10^6 active posts (groups of images and interactions) and 100×10^6 new images each day. Sizes varies considerably but an average image size approaches 100 Kbytes. [5 marks]

Answer:

The problem faced with (Instagram) was the system is extremely cache-heavy; images served from disk are served slowly and the performance of the system greatly depends on 'hot' caches. reboot meant no how caches.

Importantly the hash-tag (whats hot and whats not) are driven by cache hits so all get reset on reboot; essentially the system depends on 16 distributed Hyper systems (ridiculously large memory clusters) to front end the world.

candidates can/may include information about scale, about world latency, about frequency statistics for adjusting popularity and thus caching potential, and so forth. An interesting aspect of such image caches is the cache lasts potentially for as long as the host-cluster operates. Recovery of one cache host can be based on other "hot caches" but a cold recovery was disastrous: stressing the backend photo store, messing up the frequency counting (so the popularity-based caches were all churning until enough frequency data could be accumulated), and so forth.
